

Google

You are a Data Analyst on the Google Ads Performance team working to optimize ad campaign strategies....

Question 1: For each ad campaign segment, what are the unique ad formats used during July 2024? This will help us understand the diversity in our ad formats.

```
SELECT DISTINCT segment, ad_format
FROM fct_ad_performance AS f
INNER JOIN dim_campaign AS d
    ON f.campaign_id = d.campaign_id
WHERE campaign_date >= '2024-07-01' AND campaign_date < '2024-08-01';
```



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Question 2: How many unique campaigns had at least one rolling 7-day period in August 2024 where their total impressions exceeded 1,000? We want to identify campaigns that had a high reach in at least one 7-day window during this month.

```
with august AS(
  SELECT campaign_id, impressions, campaign_date
  FROM fct_ad_performance
  WHERE campaign_date >= '2024-08-01' AND campaign_date < '2024-09-01'),

roll7 AS(
  SELECT campaign_id,
    SUM(impressions) OVER(PARTITION BY campaign_id ORDER BY campaign_date ROWS BETWEEN 6 PRECEDING AND CURRENT ROW) AS rolling_7_day
  FROM august)

SELECT COUNT(DISTINCT(campaign_id))
FROM roll7
WHERE rolling_7_day > 1000
```



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Question 3: What is the total ROI for each campaign segment in Q3 2024? And, how does it compare to the average ROI of all campaigns (return labels 'higher than average' or 'lower than average')? We will use this to identify which segments are outperforming the average.

Note 1: ROI is defined as $(\text{revenue} - \text{cost}) / \text{cost}$.

Note 2: For average ROI across segment, calculate the ROI per segment and then calculate the average ROI across segments.

```
with q3 AS(
  SELECT *, (revenue - cost)/cost AS ROI
  FROM fct_ad_performance
  INNER JOIN dim_campaign
  ON dim_campaign.campaign_id = fct_ad_performance.campaign_id
  WHERE campaign_date >= '2024-07-01' AND campaign_date < '2024-10-01'),

roi_segment AS(
  SELECT segment,
         AVG(roi) AS avg_roi_segment,
         AVG(AVG(roi)) OVER() AS overall_avg_roi
  FROM q3
  GROUP BY segment)

SELECT *,
       CASE WHEN avg_roi_segment > overall_avg_roi THEN 'higher than average'
       ELSE 'lower than average' END AS compare_to_overall_ROI
FROM roi_segment
```

